Electric fields

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use field arrows to predict the force on a test charge.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 18.*
Bends toward the normal; frequency stays the same at a stationary boundary.

02. Worked example: explain the model
Positive charge force follows E; negative charge force opposes E.

03. Read and interpret the model
Field outward; negative test charge force toward the source.

04. Apply the physics
4 N/C right.

05. Explain your reasoning
No; it probes an existing field, assuming it does not substantially disturb the source arrangement.

06. Review: connect an earlier idea *Review: Unit 6, Day 9.*
60 m; one-way distance is half the round trip.

07. Exit check
Positive down; negative up.

08. STRETCH: test the model
Equal magnitude opposite charges, e.g.±3 C, give±12 N in this supplied field.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 18.*
Thermal peak becomes 1/3 by Wien's law; sound wavelength becomes 1/5 by λ=v/f. Different models and assumptions.

02. Worked example: explain the model
Positive charge force follows E; negative charge force opposes E.

03. Read and interpret the model
Field outward; negative test charge force toward the source.

04. Apply the physics
6.0×10^{-4} N west.

05. Explain your reasoning
No; it probes an existing field, assuming it does not substantially disturb the source arrangement.

06. Review: connect an earlier idea *Review: Unit 6, Day 9.*
λ=1.6 m; f≈212.5 Hz.

07. Exit check
One fourth; isolated point charge or spherically symmetric source outside its extent.

08. STRETCH: test the model
Magnitude 4000 N/C downward; divide force by signed charge.